

Linear Regression Model for Council District V

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City of Detroit

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Model I

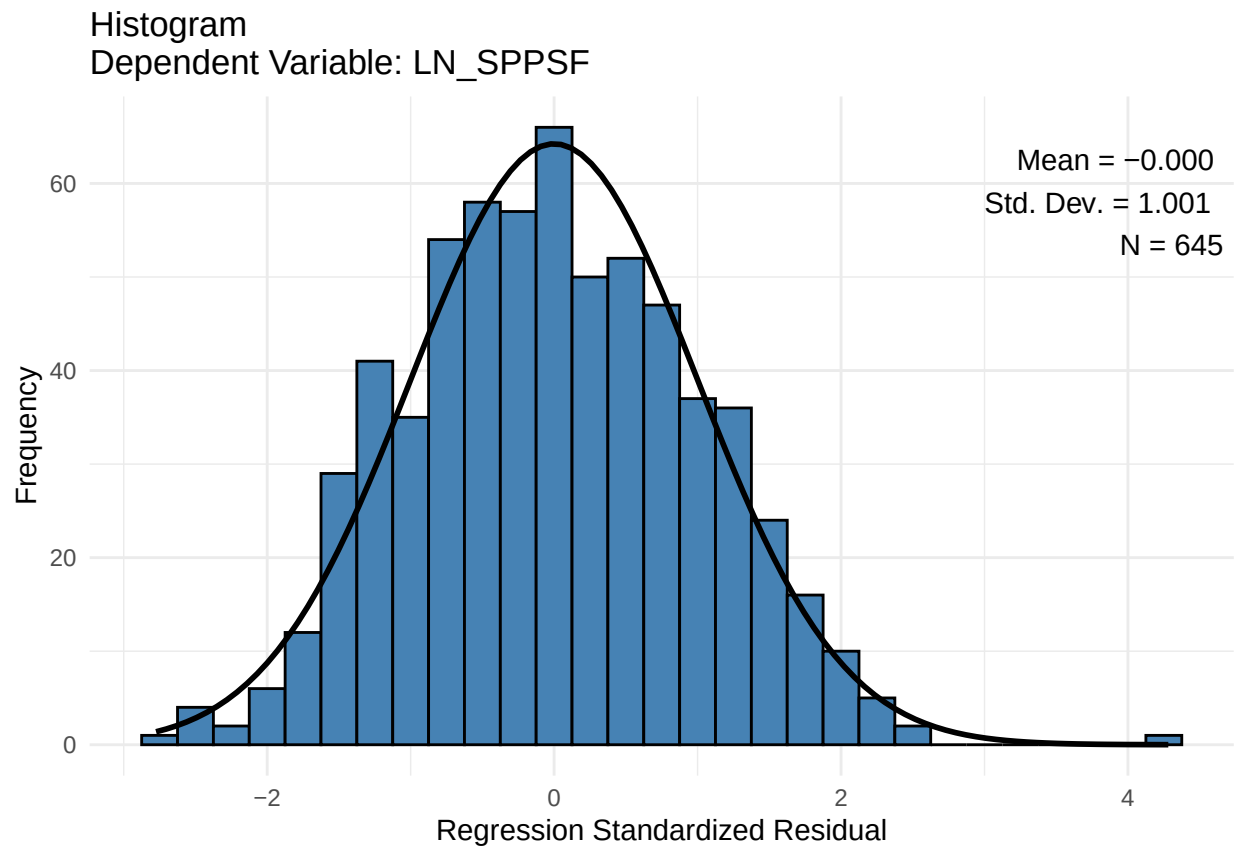


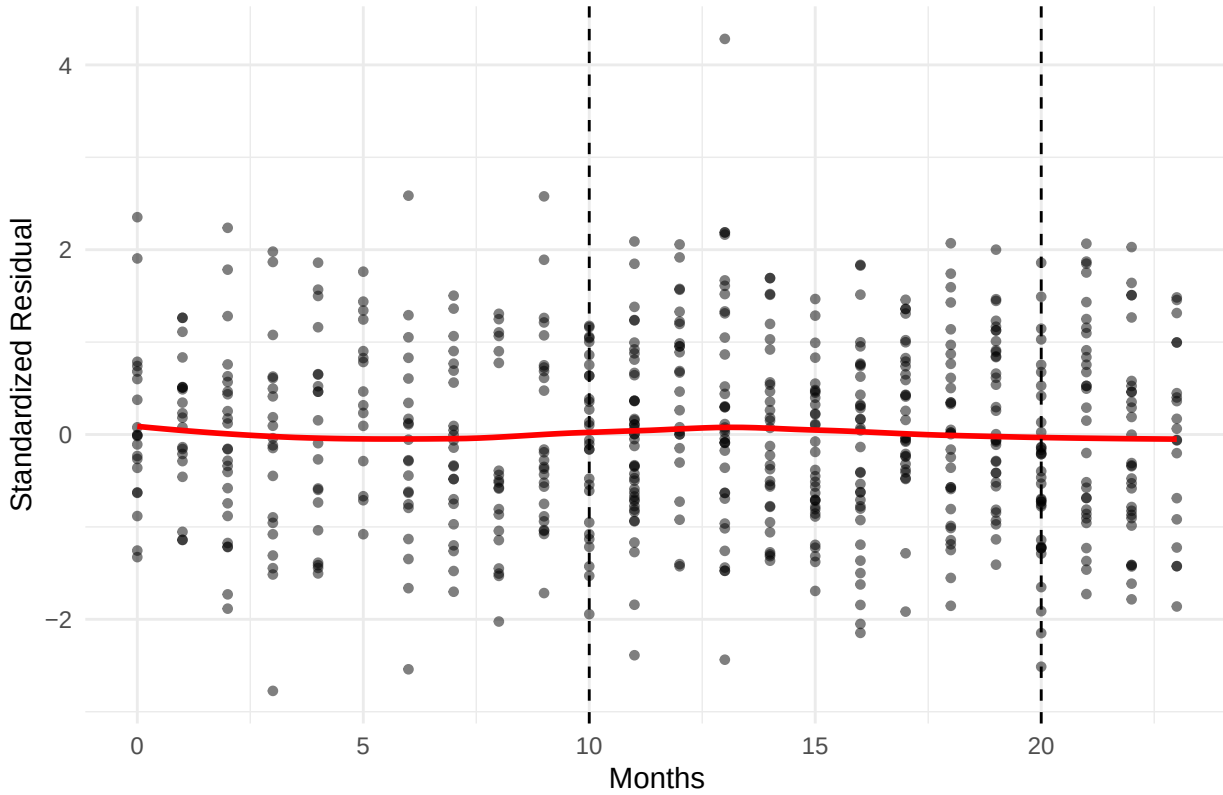
Table 1: District 5 LN_SPPSF model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
645	637	11	0.7362	0.2643	0.2563	0.7316	0.5949	32.6996	0	1445.254	1485.478

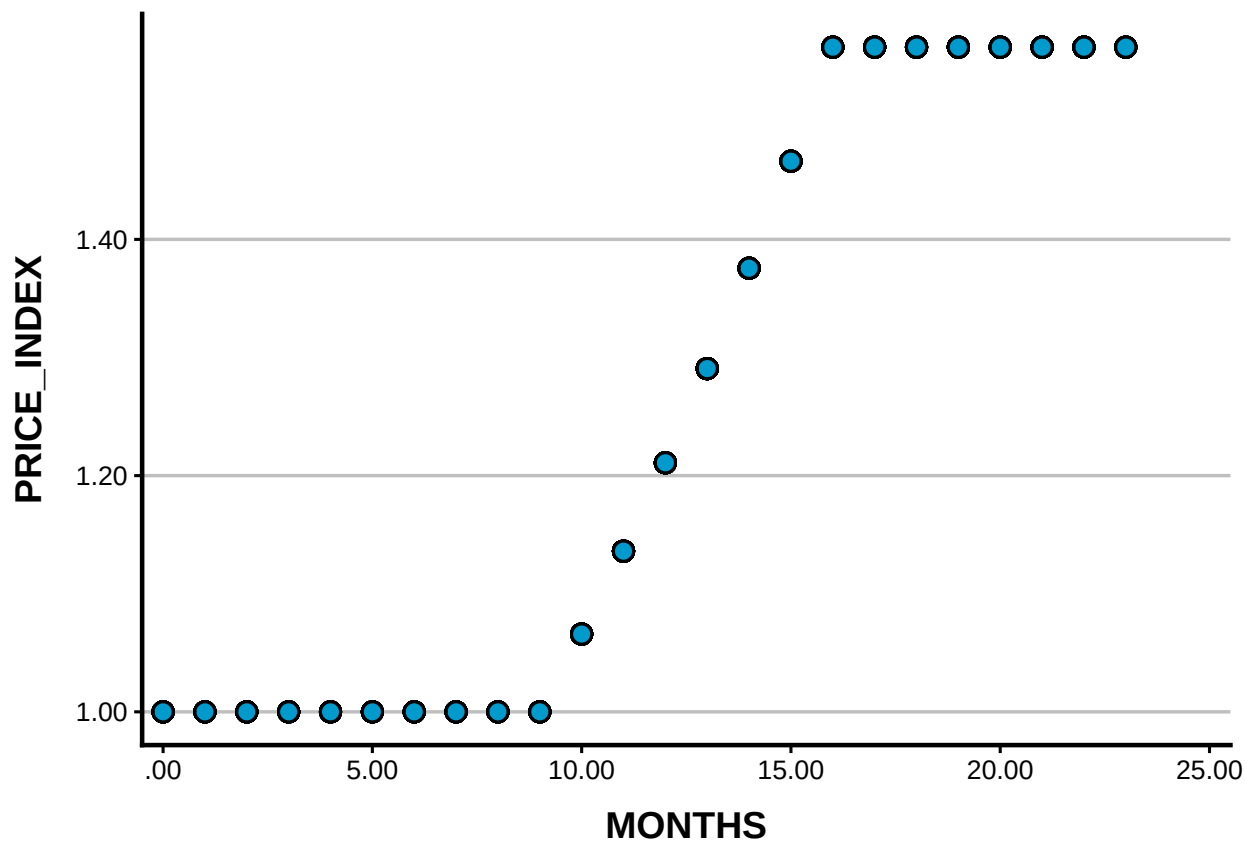
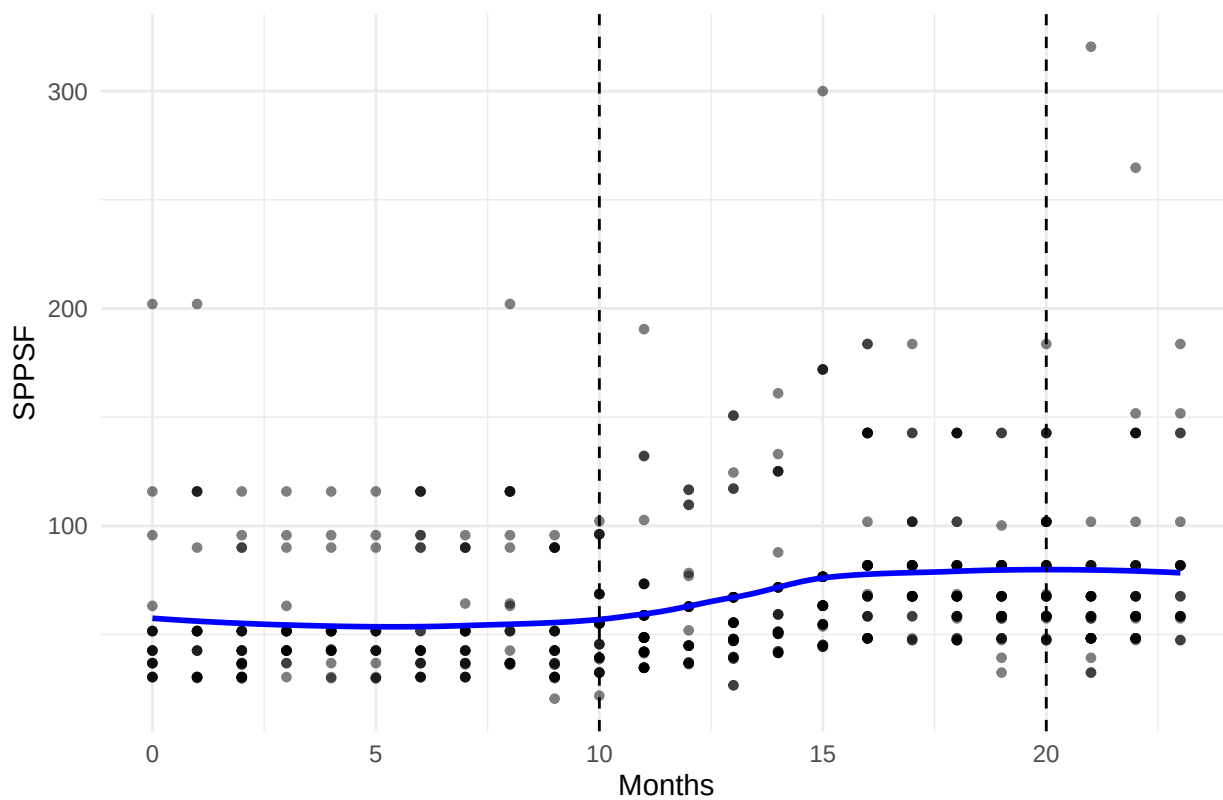
Table 2: District 5 LN_SPPSF model coefficients (95% CI)

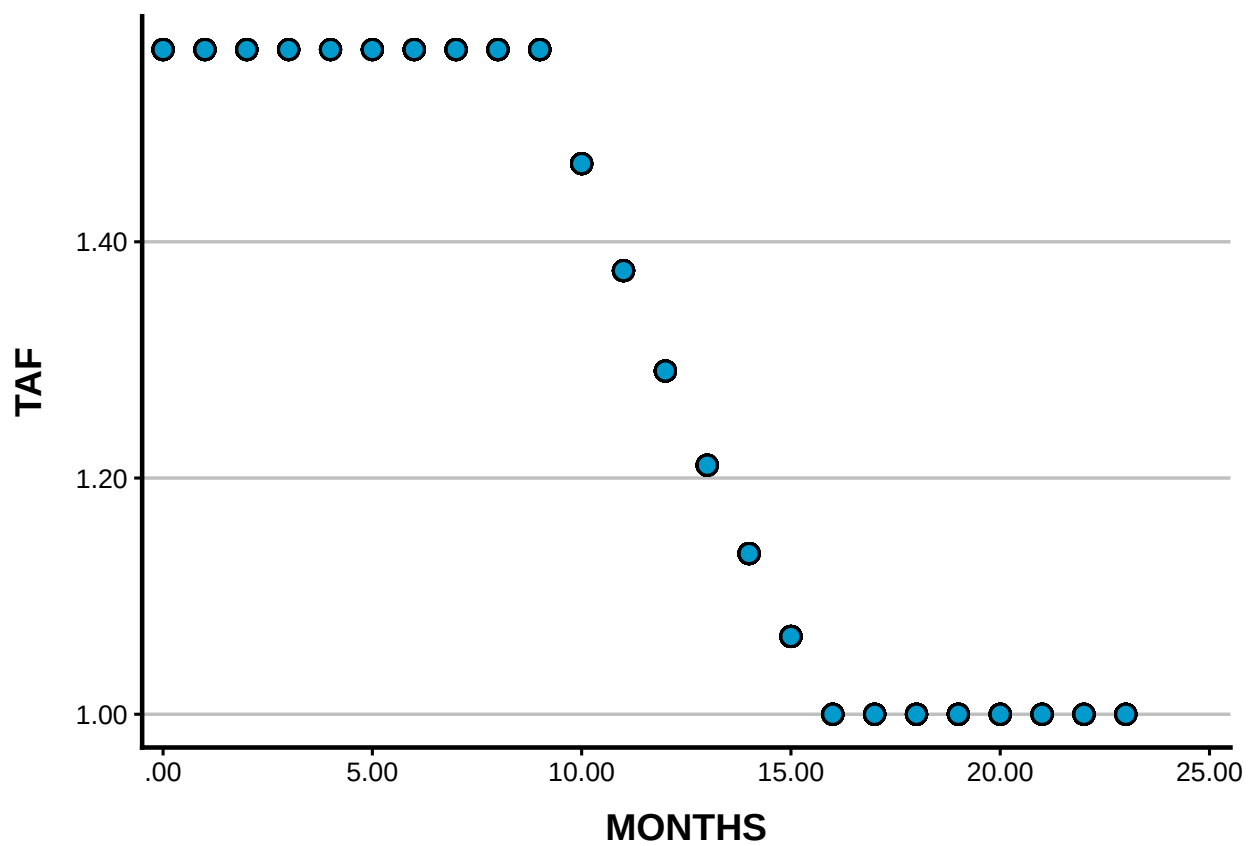
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	3.9434	62.6530	0.0629	0.0000	3.8198	4.0670
half_duplex1story	NA	NA	NA	NA	NA	NA
large	-0.1908	-3.0458	0.0627	0.0024	-0.3139	-0.0678
good	0.8087	7.2883	0.1110	0.0000	0.5908	1.0266
fair	-0.3369	-5.0686	0.0665	0.0000	-0.4675	-0.2064
poor	-0.3540	-3.4937	0.1013	0.0005	-0.5530	-0.1550
very_poor_unsound	-0.7328	-3.4964	0.2096	0.0005	-1.1443	-0.3212
qabove_avg	0.5566	6.3665	0.0874	0.0000	0.3849	0.7282
elec_baseboard	NA	NA	NA	NA	NA	NA
wall	NA	NA	NA	NA	NA	NA
months_10to16	0.0659	6.8728	0.0096	0.0000	0.0470	0.0847

Residuals for LN_SPPSF Model



Months vs Predicted SPPSF – LN_SPPSF Model





Model II

Histogram
Dependent Variable: LN_PRICE (No ECFs)

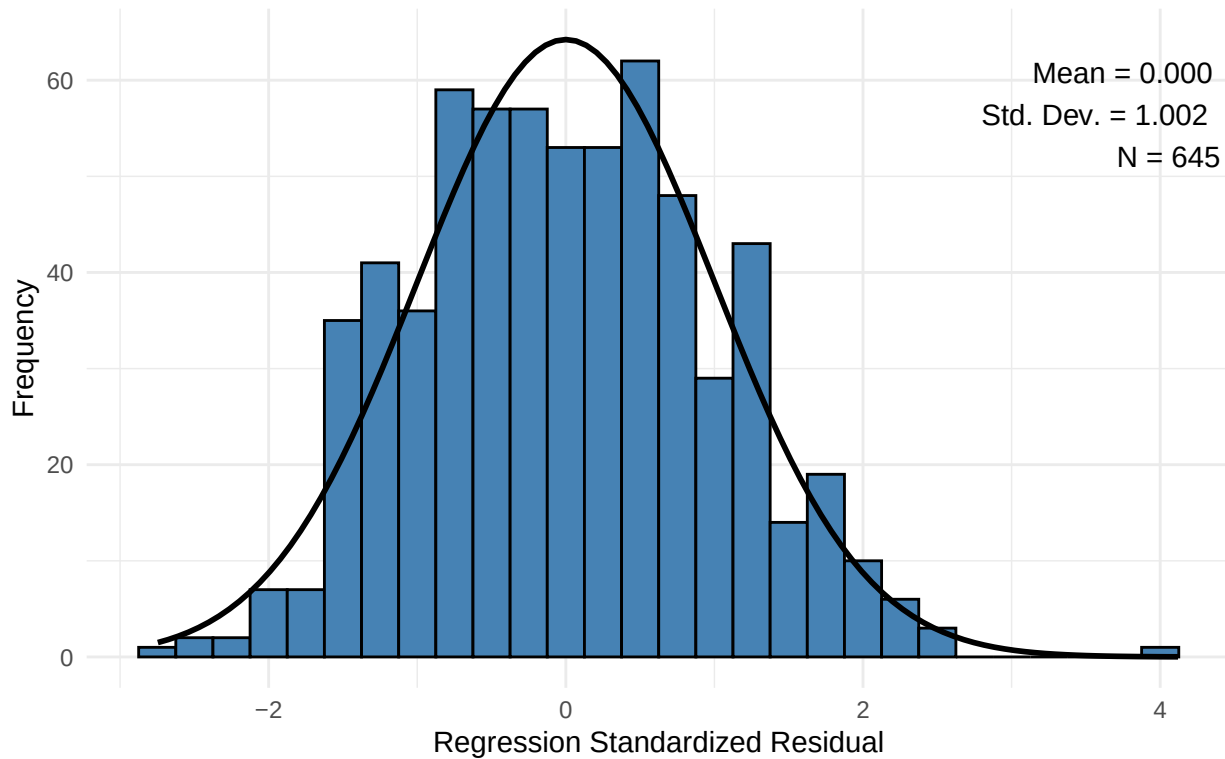


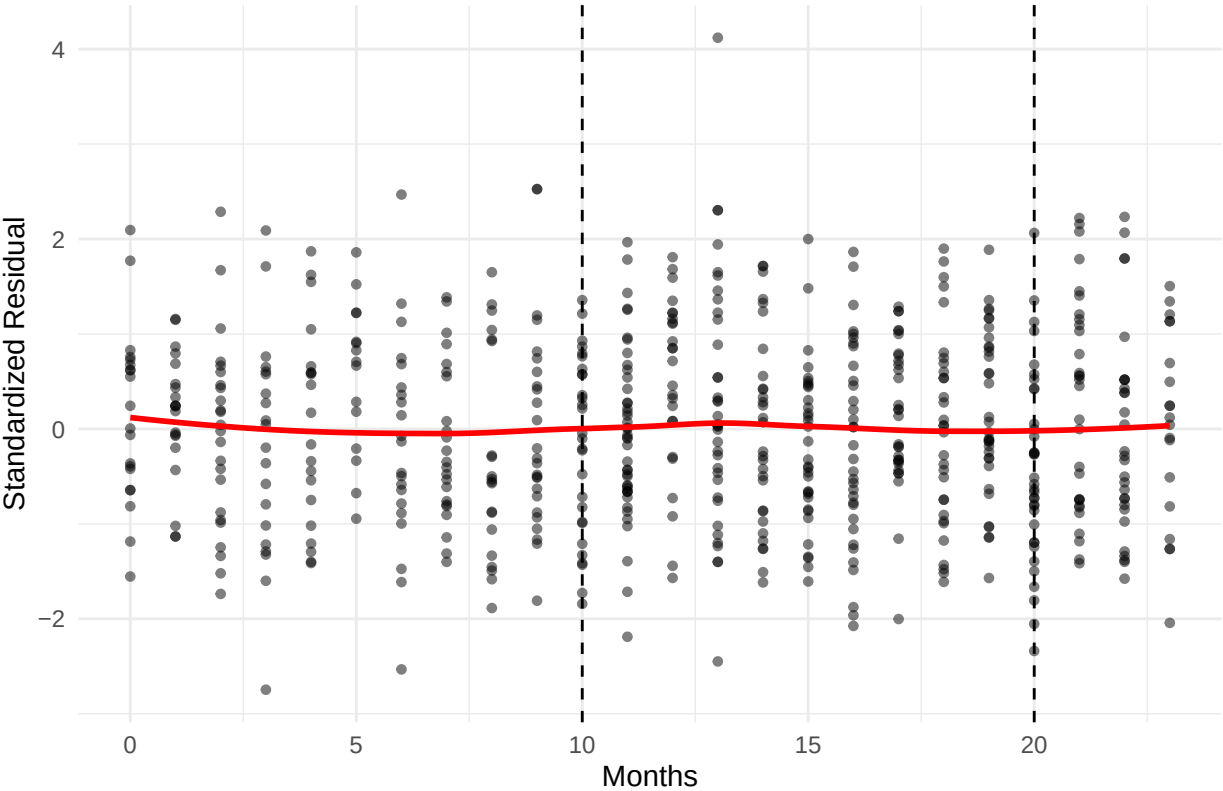
Table 3: District 5 LN_PRICE (No ECFs) model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
645	631	17	0.7297	0.4286	0.4168	0.7217	0.5923	36.4013	0	1439.791	1506.83

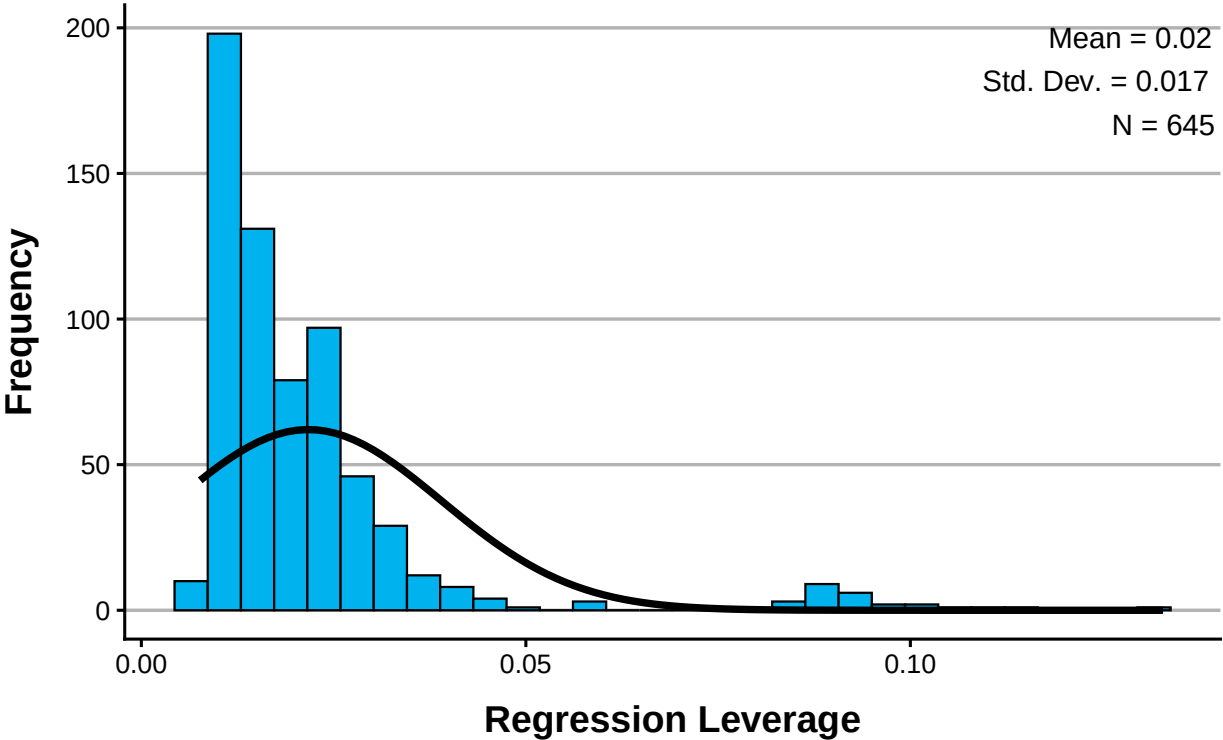
Table 4: District 5 LN_PRICE (No ECFs) model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.8649	124.6204	0.0872	0.0000	10.6937	11.0361
ln_landratio	0.2792	2.6489	0.1054	0.0083	0.0722	0.4862
ln_base_bldgsize_ratio	0.8703	5.1831	0.1679	0.0000	0.5406	1.2001
arterial	-0.1648	-1.9639	0.0839	0.0500	-0.3297	0.0000
single_family1story	0.1173	1.8546	0.0633	0.0641	-0.0069	0.2415
half_duplex1story	NA	NA	NA	NA	NA	NA
largest	0.1921	2.0792	0.0924	0.0380	0.0107	0.3735
good	0.8355	7.5549	0.1106	0.0000	0.6183	1.0527
fair	-0.3424	-5.1486	0.0665	0.0000	-0.4731	-0.2118
poor	-0.3677	-3.6491	0.1008	0.0003	-0.5655	-0.1698
very_poor_unsound	-0.7412	-3.5616	0.2081	0.0004	-1.1499	-0.3325
qabove_avg	0.5242	4.9284	0.1064	0.0000	0.3153	0.7331
yb_post_1961	-0.4572	-2.1061	0.2171	0.0356	-0.8835	-0.0309
fireplace	-0.1264	-1.9297	0.0655	0.0541	-0.2550	0.0022
elec_baseboard	NA	NA	NA	NA	NA	NA
wall	NA	NA	NA	NA	NA	NA
months_10to16	0.0650	6.5612	0.0099	0.0000	0.0455	0.0844

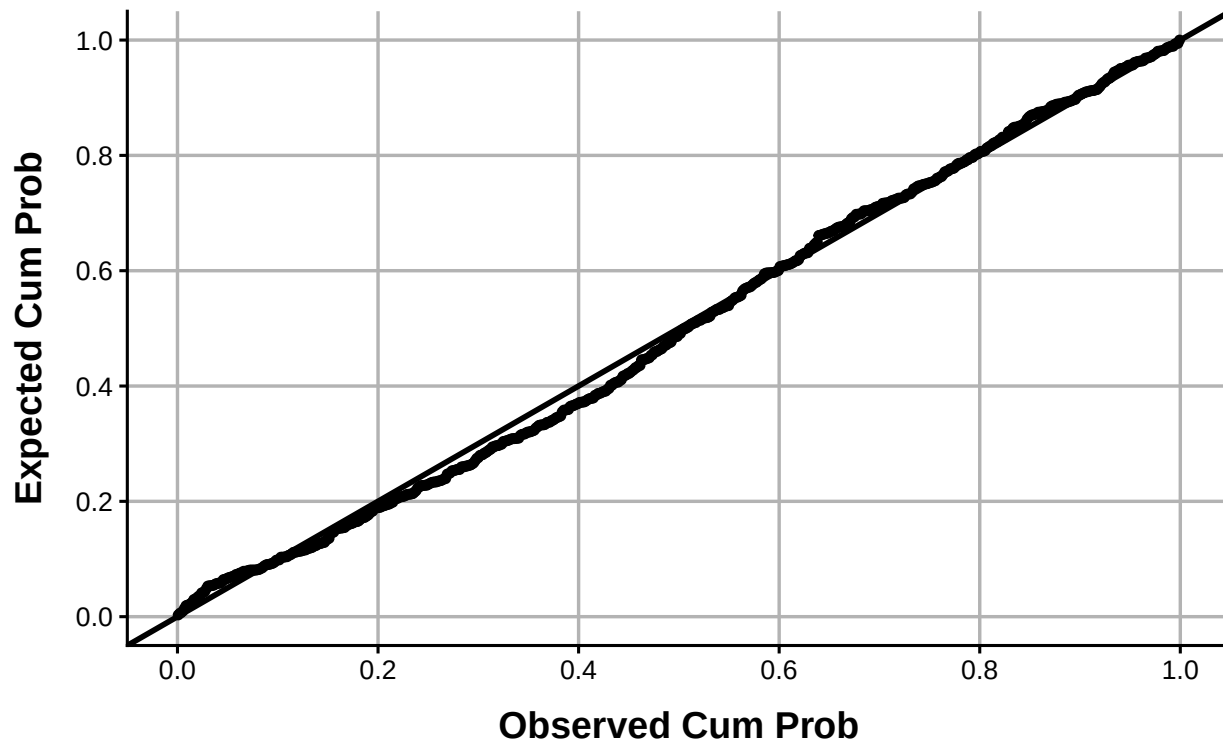
Residuals for LN_PRICE (No ECFs) Model



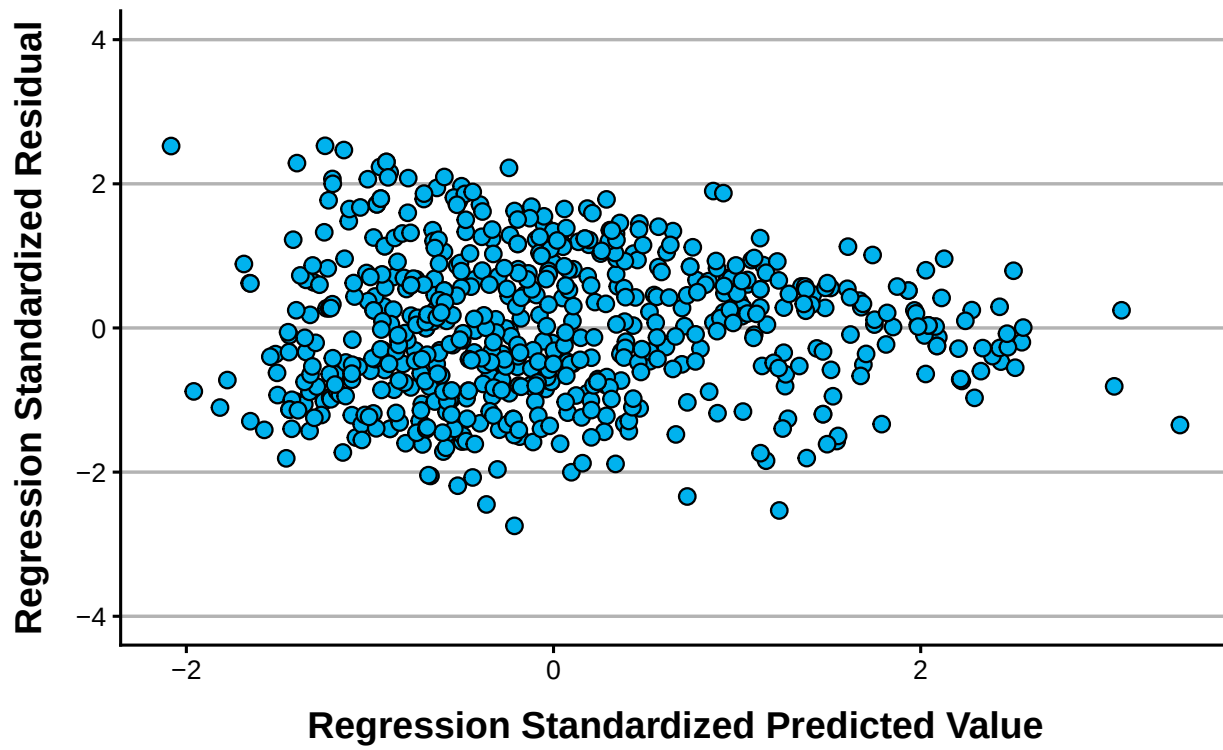
Histogram
Dependent Variable: LN\ PRICE

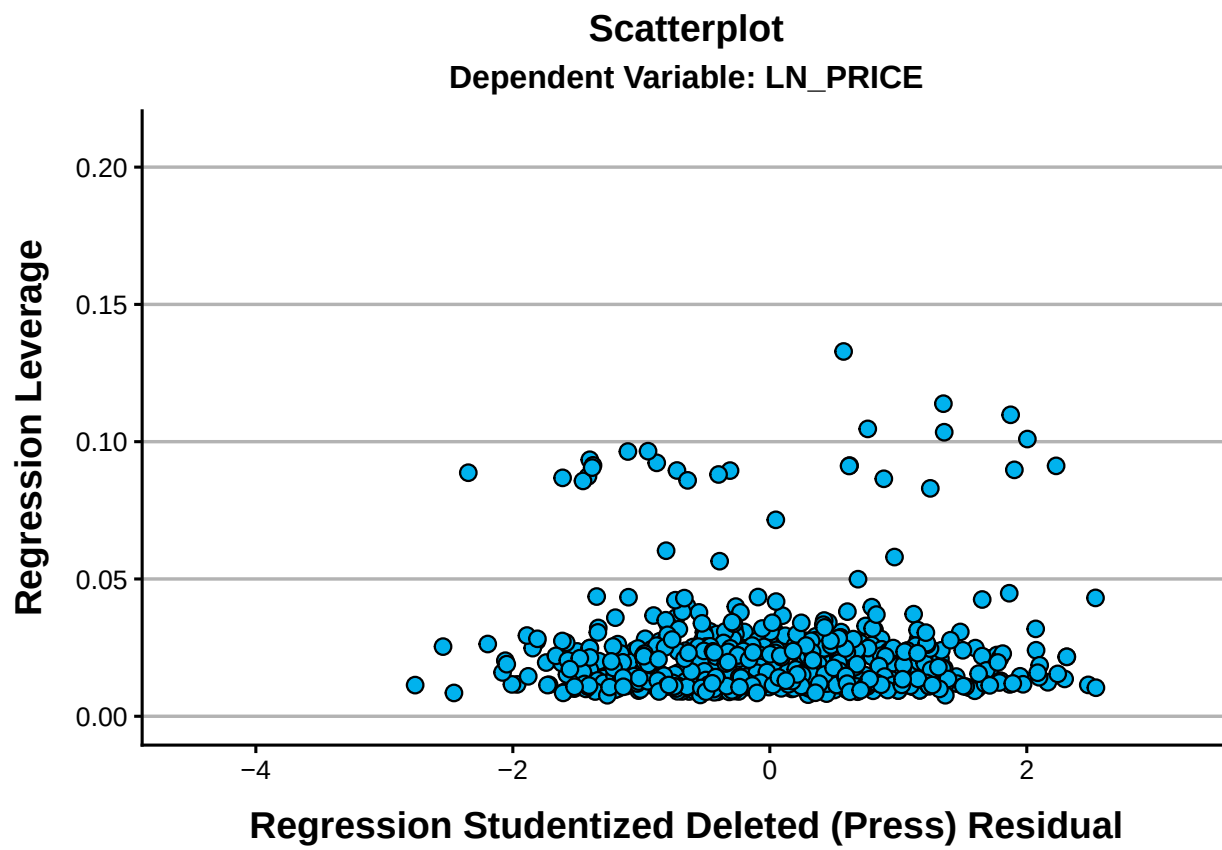
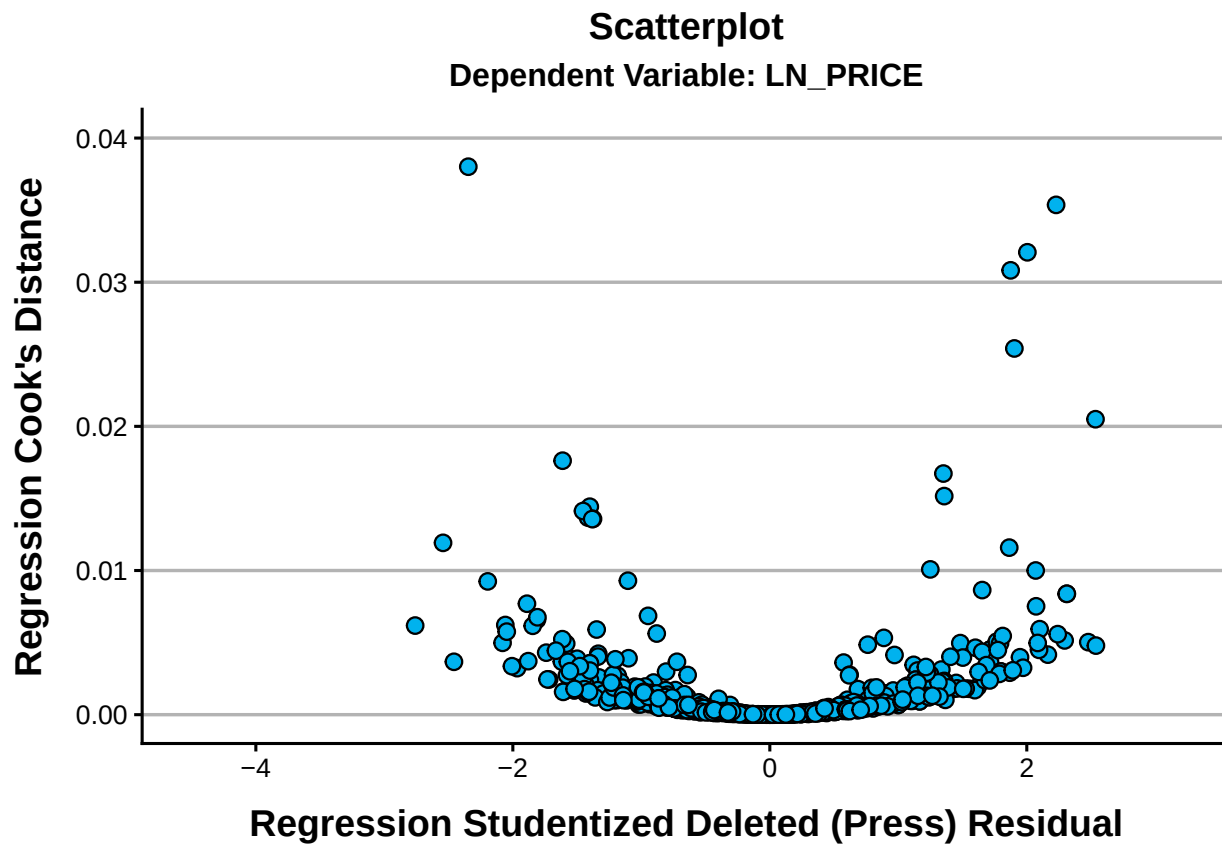


Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE





Model III

Initial Input

Table 5: LN_PRICE model (base) assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
645	1.023324	1.206802	0.8765876	1.376705	54.05011	-0.5166936

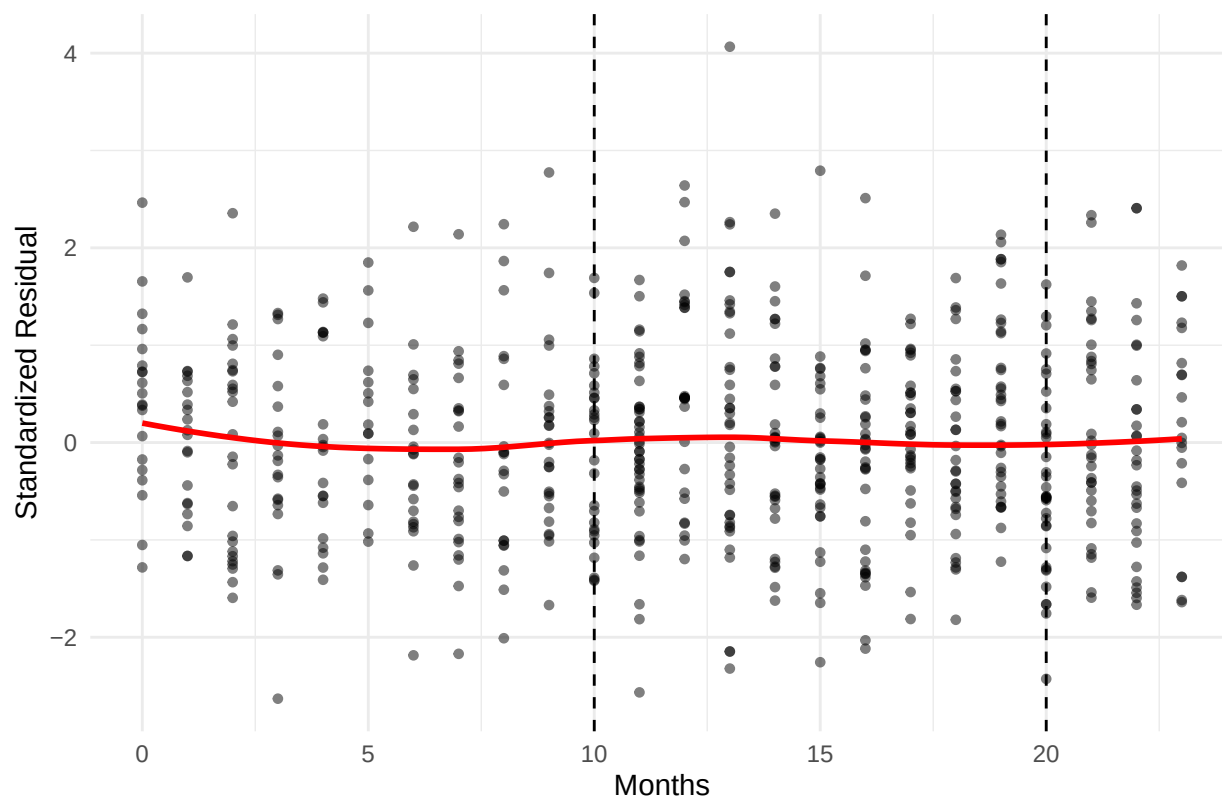
Table 6: District 5 LN_PRICE model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
645	615	33	0.6471	0.562	0.5413	0.6319	0.5022	27.2109	0	1300.25	1438.797

Table 7: District 5 LN_PRICE model coefficients (95% CI)

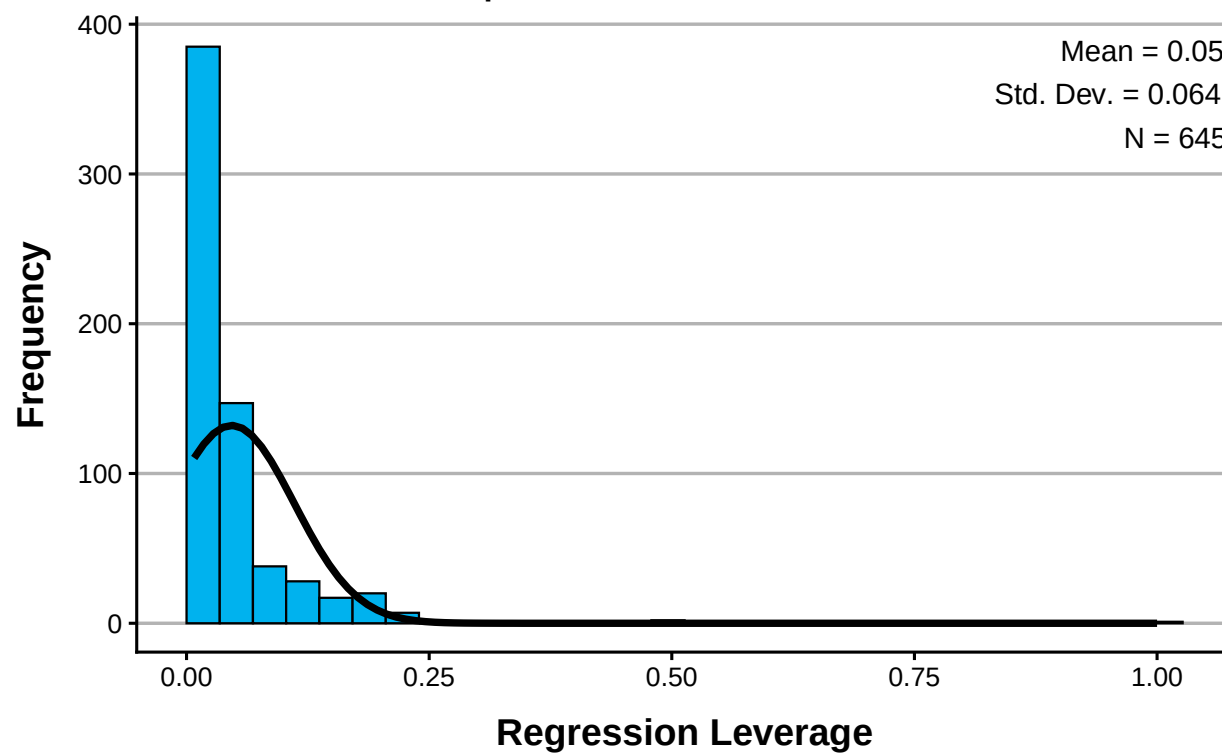
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.0523	42.6025	0.2594	0.0000	10.5429	11.5618
ln_landratio	0.2549	2.6615	0.0958	0.0080	0.0668	0.4430
ln_base_bldgsize_ratio	0.7678	6.1998	0.1238	0.0000	0.5246	1.0111
single_family1story	0.1329	2.3305	0.0570	0.0201	0.0209	0.2449
half_duplex1story	NA	NA	NA	NA	NA	NA
good	0.8041	8.1279	0.0989	0.0000	0.6098	0.9984
fair	-0.3359	-5.6346	0.0596	0.0000	-0.4529	-0.2188
poor	-0.4601	-5.0492	0.0911	0.0000	-0.6391	-0.2812
very_poor_unsound	-0.8632	-4.2791	0.2017	0.0000	-1.2594	-0.4671
qabove_avg	0.4232	3.7258	0.1136	0.0002	0.2001	0.6462
qbelow_avg	-0.2079	-1.7648	0.1178	0.0781	-0.4393	0.0234
yb_pre_1929	-0.4579	-1.8144	0.2523	0.0701	-0.9534	0.0377
yb_post_1961	-0.7401	-2.3330	0.3172	0.0200	-1.3630	-0.1171
ln_garagespaces	0.1335	2.1488	0.0621	0.0320	0.0115	0.2555
elec_baseboard	NA	NA	NA	NA	NA	NA
wall	NA	NA	NA	NA	NA	NA
ecf_5R503	0.6533	2.9946	0.2182	0.0029	0.2249	1.0818
ecf_5R504	0.5795	4.3303	0.1338	0.0000	0.3167	0.8423
ecf_5R506	0.4838	1.7087	0.2831	0.0880	-0.0722	1.0398
ecf_5R509	0.8107	3.6714	0.2208	0.0003	0.3771	1.2444
ecf_5R511	0.6250	2.8310	0.2208	0.0048	0.1914	1.0585
ecf_5R512	0.6601	3.1552	0.2092	0.0017	0.2493	1.0710
ecf_5R514	0.7685	7.8183	0.0983	0.0000	0.5754	0.9615
ecf_5R516	1.1802	1.7297	0.6823	0.0842	-0.1597	2.5201
ecf_5R517	1.3966	4.7471	0.2942	0.0000	0.8189	1.9744
ecf_5R521	0.6108	3.8219	0.1598	0.0001	0.2970	0.9247
ecf_5R524	1.9523	3.8401	0.5084	0.0001	0.9539	2.9507
ecf_5R529	0.8362	8.9061	0.0939	0.0000	0.6518	1.0206
ecf_5R530	0.9897	3.6705	0.2696	0.0003	0.4602	1.5193
ecf_5R531	0.2761	2.1739	0.1270	0.0301	0.0267	0.5255
ecf_5R532	0.5558	1.7869	0.3110	0.0744	-0.0550	1.1666
ecf_5R539	0.4277	1.6528	0.2588	0.0989	-0.0805	0.9359
months_10to16	0.0674	7.5870	0.0089	0.0000	0.0500	0.0849

Residuals for LN_PRICE Model

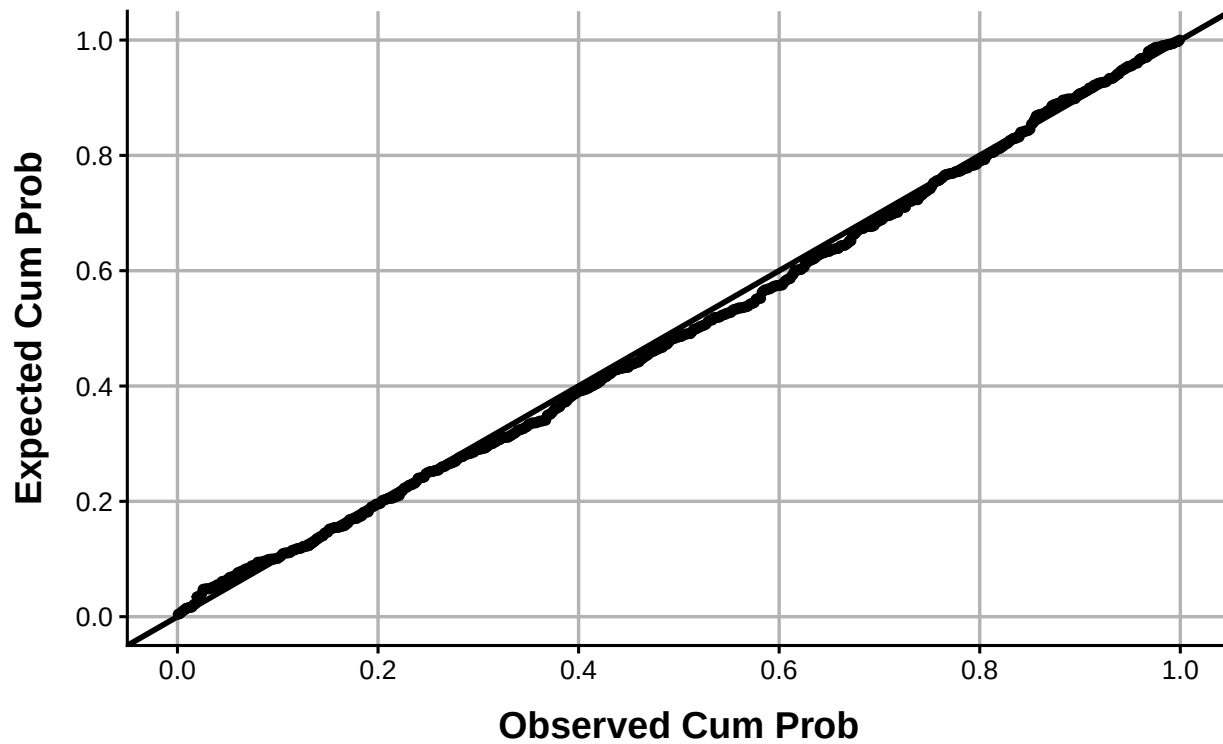


Histogram

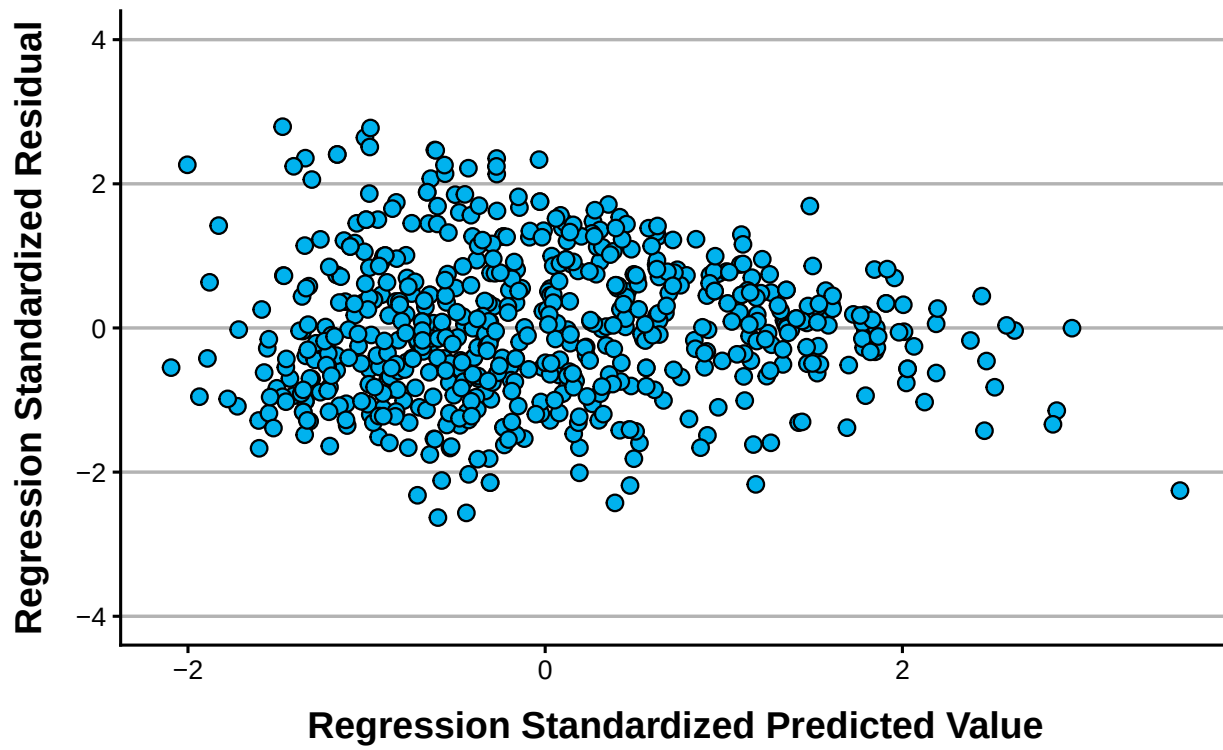
Dependent Variable: LN\ PRICE

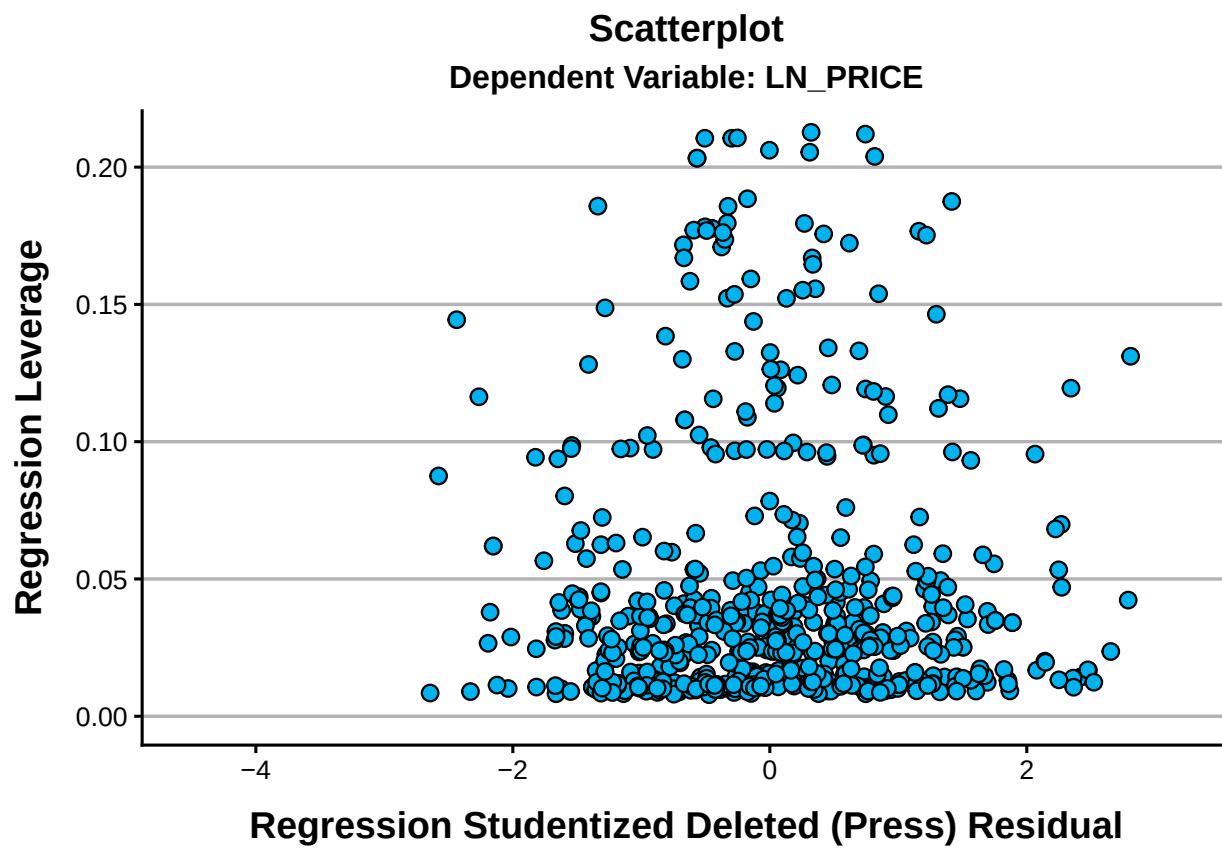
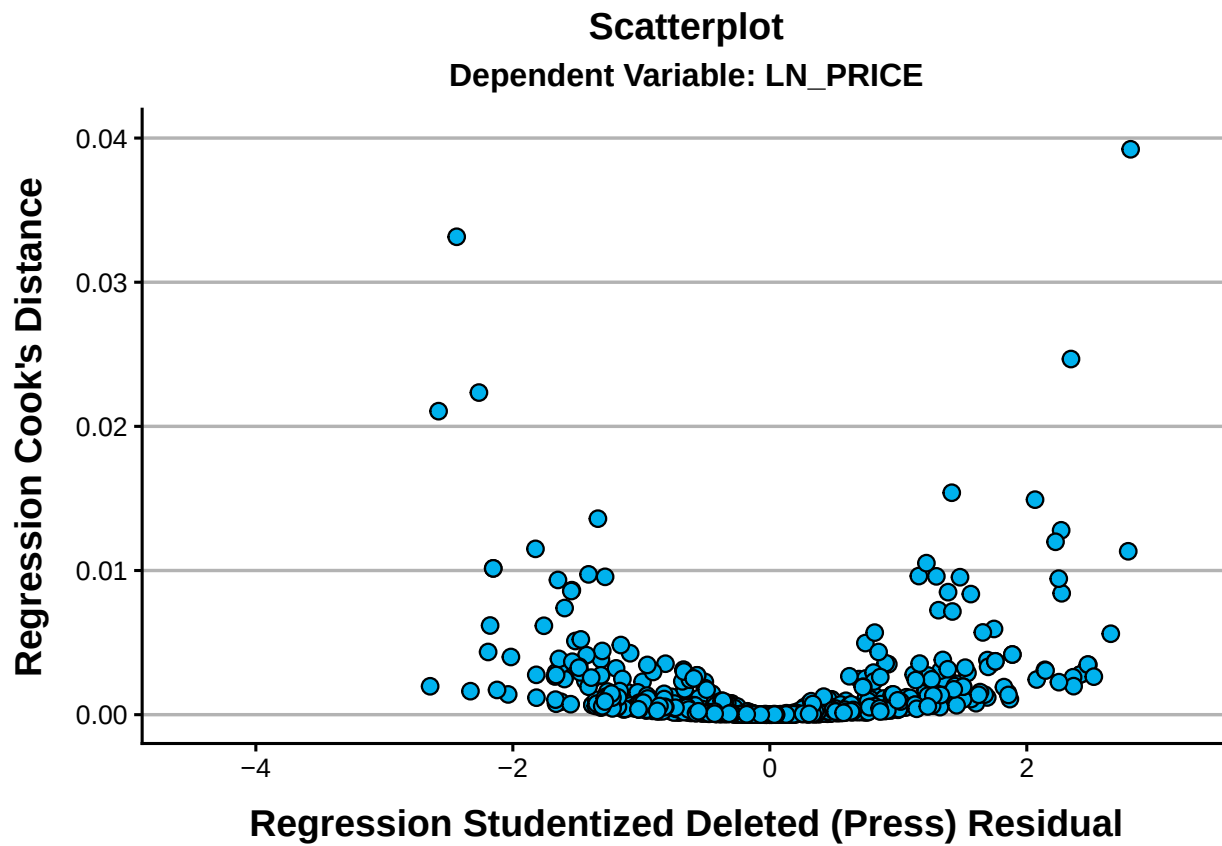


Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE





Filtered Input

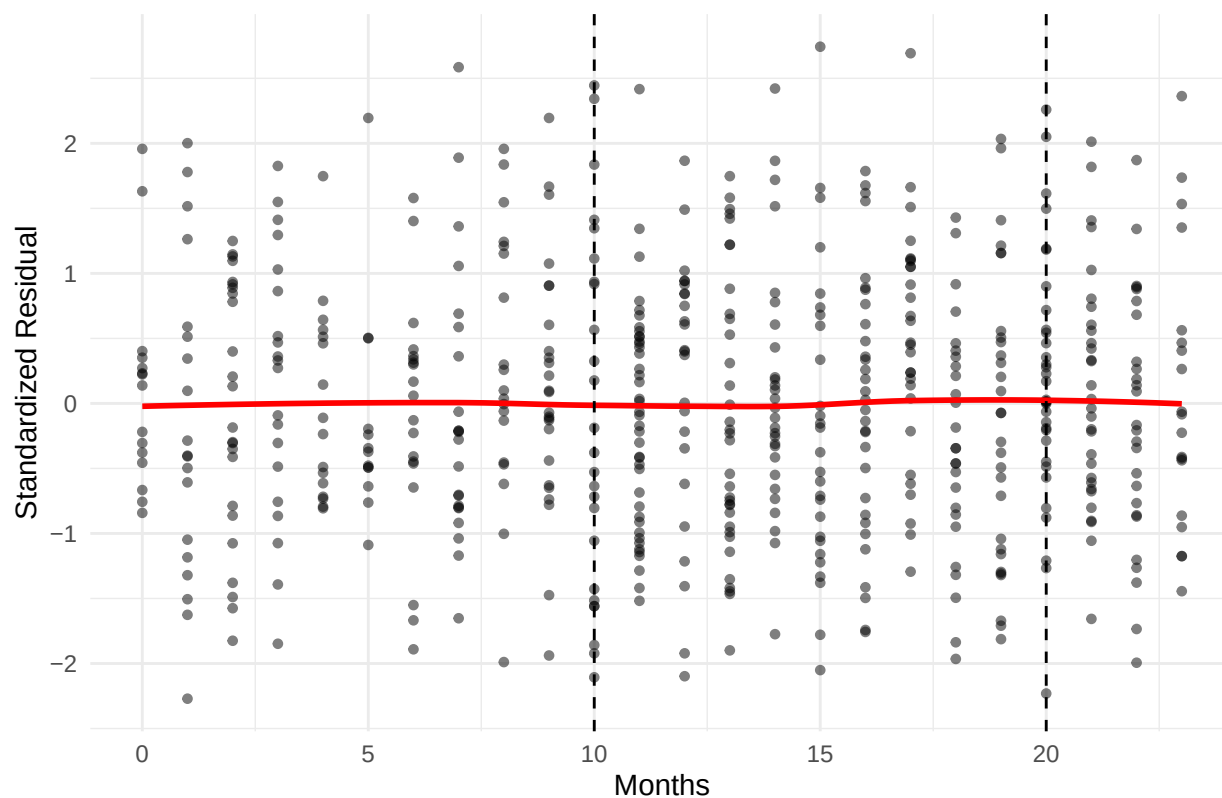
Table 8: District 5 LN_PRICE (Inliers) model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
594	559	40	0.5331	0.6896	0.6707	0.5171	0.4174	36.5292	0	974.2131	1132.141

Table 9: District 5 LN_PRICE (Inliers) model coefficients (95% CI)

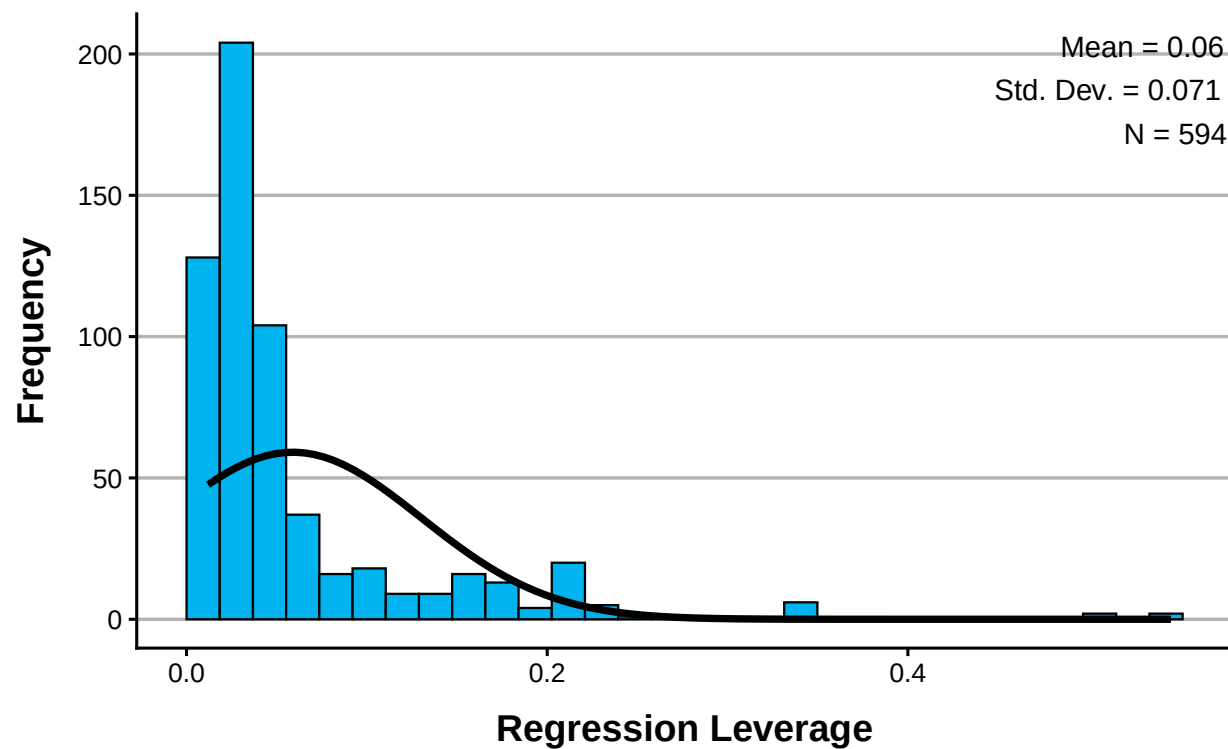
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.5587	123.9813	0.0852	0.0000	10.3914	10.7260
ln_base_bldgsize_ratio	0.6124	4.3707	0.1401	0.0000	0.3372	0.8876
single_family1story	0.1346	2.6885	0.0501	0.0074	0.0363	0.2329
half_duplex1story	NA	NA	NA	NA	NA	NA
half_duplex2story	NA	NA	NA	NA	NA	NA
smallest	-0.4576	-1.8860	0.2426	0.0598	-0.9341	0.0190
small	-0.3762	-1.9172	0.1962	0.0557	-0.7617	0.0092
large	-0.0940	-1.6841	0.0558	0.0927	-0.2037	0.0156
two_plus_baths	0.1468	1.7836	0.0823	0.0750	-0.0149	0.3085
good	0.9321	10.6079	0.0879	0.0000	0.7595	1.1047
fair	-0.4450	-8.7119	0.0511	0.0000	-0.5453	-0.3447
poor	-0.5599	-7.0823	0.0791	0.0000	-0.7152	-0.4046
very_poor_unsound	-1.2166	-5.6623	0.2149	0.0000	-1.6387	-0.7946
qabove_avg	0.2551	2.3596	0.1081	0.0186	0.0427	0.4674
ln_garagespaces	0.1638	3.0699	0.0534	0.0022	0.0590	0.2686
fireplace	0.1012	1.8706	0.0541	0.0619	-0.0051	0.2074
elec_baseboard	NA	NA	NA	NA	NA	NA
wall	NA	NA	NA	NA	NA	NA
ecf_5R503	1.0676	5.3998	0.1977	0.0000	0.6792	1.4559
ecf_5R504	0.7154	5.8993	0.1213	0.0000	0.4772	0.9536
ecf_5R506	0.4853	1.8553	0.2616	0.0641	-0.0285	0.9990
ecf_5R507	0.2238	1.6806	0.1331	0.0934	-0.0378	0.4853
ecf_5R508	0.7737	2.3584	0.3281	0.0187	0.1293	1.4181
ecf_5R509	0.9737	4.9913	0.1951	0.0000	0.5905	1.3568
ecf_5R511	0.6436	3.1091	0.2070	0.0020	0.2370	1.0502
ecf_5R512	0.7298	3.9457	0.1850	0.0001	0.3665	1.0931
ecf_5R513	0.6047	2.5446	0.2376	0.0112	0.1379	1.0714
ecf_5R514	0.8631	10.1827	0.0848	0.0000	0.6966	1.0296
ecf_5R516	NA	NA	NA	NA	NA	NA
ecf_5R517	1.5094	6.1561	0.2452	0.0000	1.0278	1.9910
ecf_5R518	0.5688	1.7341	0.3280	0.0835	-0.0755	1.2130
ecf_5R521	0.4506	2.8001	0.1609	0.0053	0.1345	0.7667
ecf_5R524	2.6431	5.9255	0.4461	0.0000	1.7669	3.5193
ecf_5R529	0.9289	11.5989	0.0801	0.0000	0.7716	1.0863
ecf_5R530	0.9141	3.7293	0.2451	0.0002	0.4326	1.3956
ecf_5R531	0.3597	3.3103	0.1086	0.0010	0.1463	0.5731
ecf_5R532	0.9810	3.6749	0.2670	0.0003	0.4567	1.5054
ecf_5R535	0.8491	2.0925	0.4058	0.0368	0.0521	1.6462
ecf_5R539	0.4199	1.9603	0.2142	0.0505	-0.0008	0.8407
months_10to16	0.0691	9.0807	0.0076	0.0000	0.0541	0.0840

Residuals for LN_PRICE (Inliers) Model



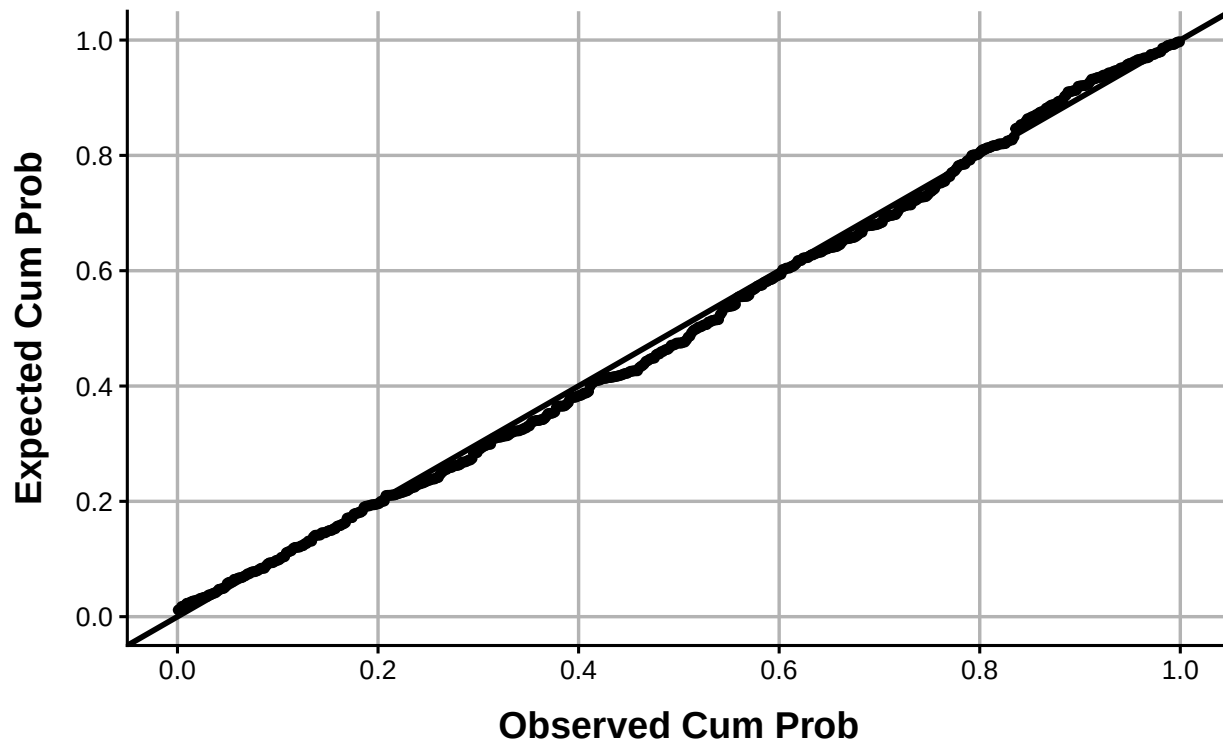
Histogram

Dependent Variable: LN\ PRICE



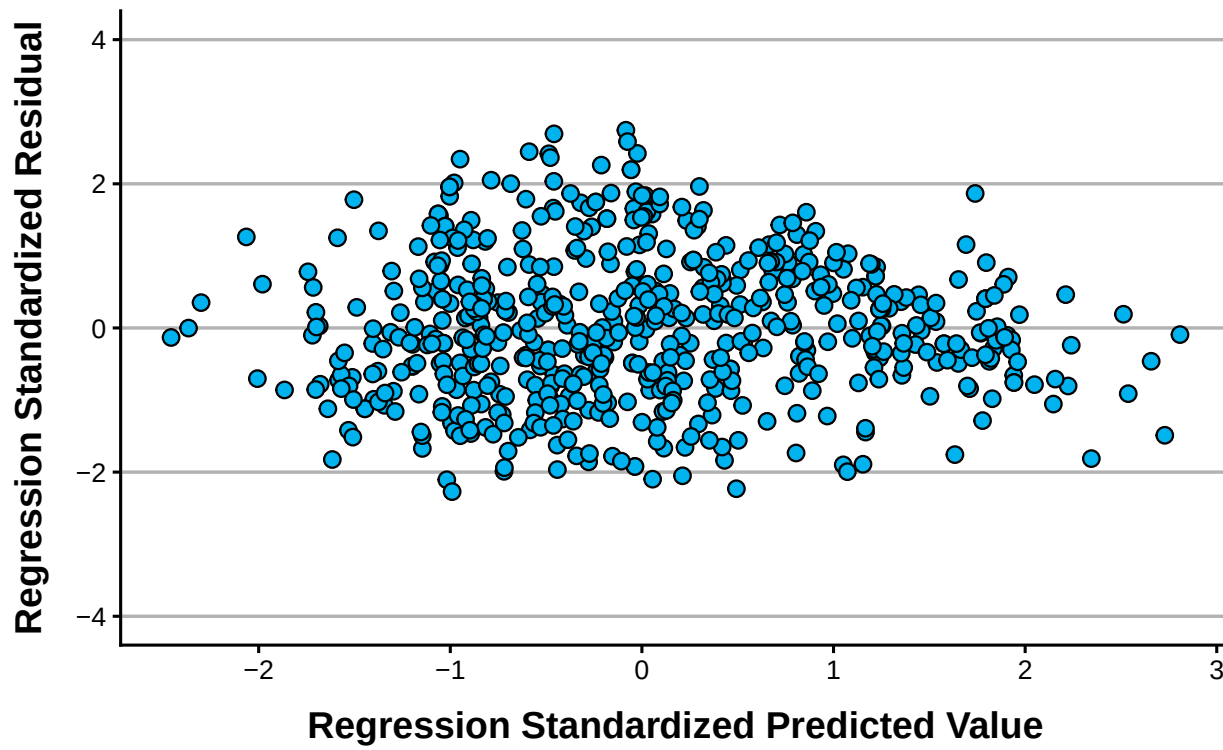
Normal P-P Plot of Regression Standardized Residual

Dependent Variable: LN_PRICE

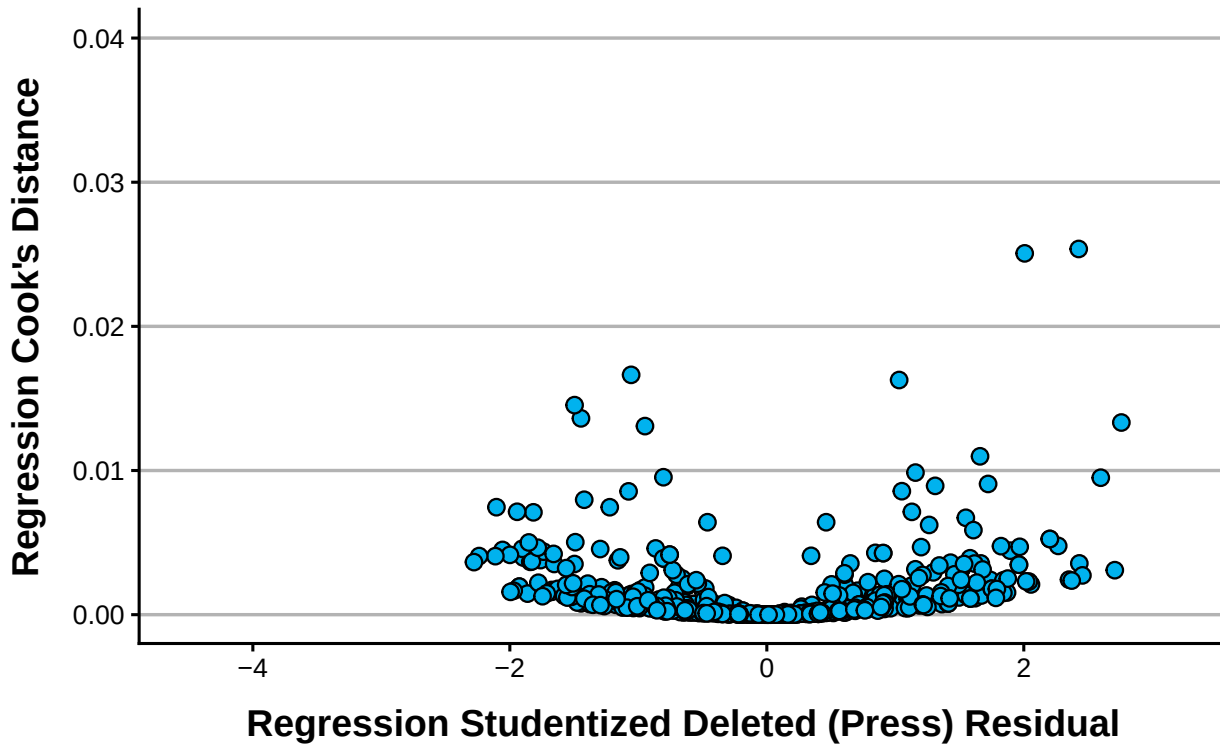


Scatterplot

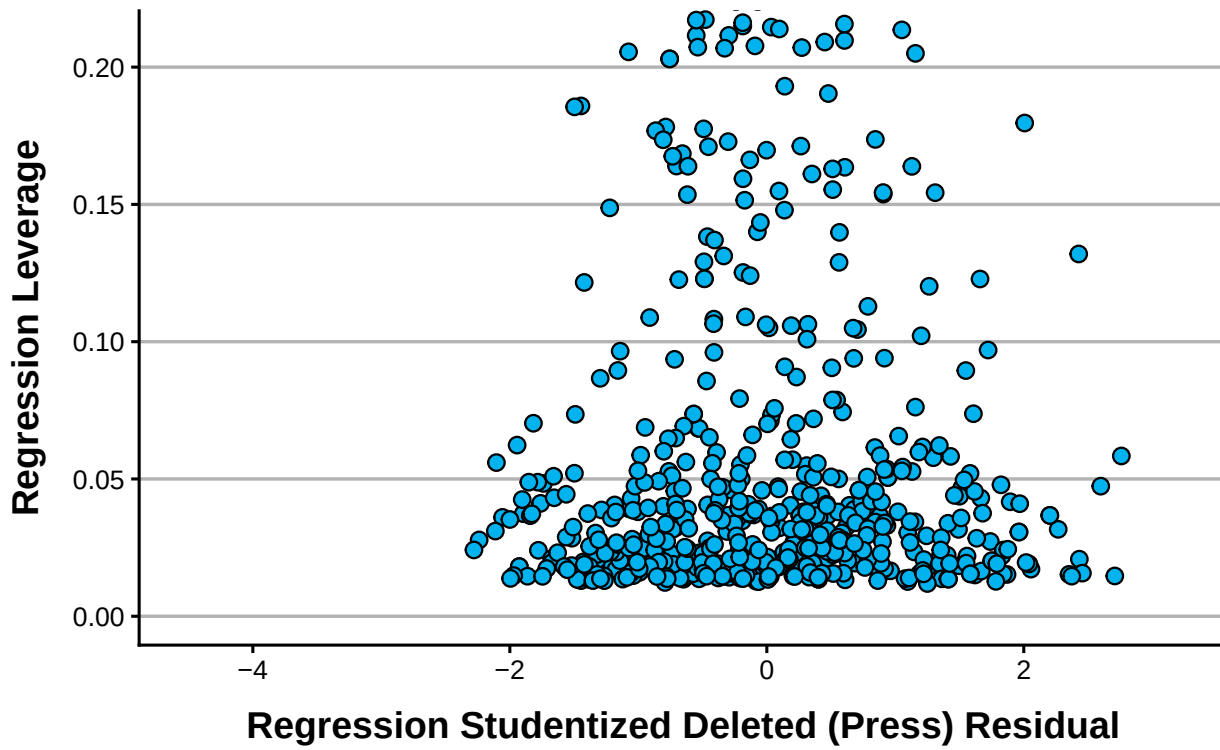
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE



Model IV

Table 10: District 5 LN_TASP Model I assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
645	1.260879	1.48448	1.086465	1.366339	54.7749	-0.6329415

Table 11: District 5 LN_TASP Model II assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
645	1.249726	1.488287	1.084509	1.372314	55.69024	-0.6423676

Table 12: District 5 LN_TASP Model I fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
594	560	39	0.5327	0.6775	0.6585	0.5172	0.4169	35.6448	0	972.4817	1126.023

Table 13: District 5 LN_TASP Model I coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.0267	134.3550	0.0821	0.0000	10.8655	11.1879
ln_base_bldgsize_ratio	0.6117	4.3691	0.1400	0.0000	0.3367	0.8867
single_family1story	0.1414	2.9369	0.0481	0.0035	0.0468	0.2360
half_duplex1story	NA	NA	NA	NA	NA	NA
half_duplex2story	NA	NA	NA	NA	NA	NA
smallest	-0.4524	-1.8675	0.2422	0.0623	-0.9282	0.0234
small	-0.3825	-1.9546	0.1957	0.0511	-0.7669	0.0019
large	-0.0942	-1.6887	0.0558	0.0918	-0.2038	0.0154
two_plus_baths	0.1466	1.7827	0.0823	0.0752	-0.0149	0.3082
good	0.9285	10.6091	0.0875	0.0000	0.7566	1.1004
fair	-0.4439	-8.7041	0.0510	0.0000	-0.5440	-0.3437
poor	-0.5573	-7.0692	0.0788	0.0000	-0.7121	-0.4024
very_poor_unsound	-1.2130	-5.6523	0.2146	0.0000	-1.6345	-0.7915
qabove_avg	0.2569	2.3797	0.1080	0.0177	0.0449	0.4690
ln_garagespaces	0.1624	3.0504	0.0533	0.0024	0.0578	0.2670
fireplace	0.1020	1.8881	0.0540	0.0595	-0.0041	0.2081
elec_baseboard	NA	NA	NA	NA	NA	NA
wall	NA	NA	NA	NA	NA	NA
ecf_5R503	1.0670	5.4006	0.1976	0.0000	0.6789	1.4551
ecf_5R504	0.7185	5.9360	0.1210	0.0000	0.4807	0.9562
ecf_5R506	0.4759	1.8252	0.2607	0.0685	-0.0362	0.9879
ecf_5R507	0.2245	1.6874	0.1330	0.0921	-0.0368	0.4858
ecf_5R508	0.7793	2.3783	0.3277	0.0177	0.1357	1.4228
ecf_5R509	0.9769	5.0139	0.1948	0.0000	0.5942	1.3596
ecf_5R511	0.6442	3.1138	0.2069	0.0019	0.2378	1.0505
ecf_5R512	0.7337	3.9732	0.1847	0.0001	0.3710	1.0965
ecf_5R513	0.6002	2.5290	0.2373	0.0117	0.1340	1.0663
ecf_5R514	0.8634	10.1928	0.0847	0.0000	0.6970	1.0297
ecf_5R516	NA	NA	NA	NA	NA	NA
ecf_5R517	1.5035	6.1432	0.2447	0.0000	1.0228	1.9842
ecf_5R518	0.5723	1.7463	0.3277	0.0813	-0.0714	1.2159
ecf_5R521	0.4486	2.7902	0.1608	0.0054	0.1328	0.7644
ecf_5R524	2.6367	5.9175	0.4456	0.0000	1.7615	3.5119

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_5R529	0.9286	11.6025	0.0800	0.0000	0.7714	1.0858
ecf_5R530	0.9185	3.7518	0.2448	0.0002	0.4376	1.3993
ecf_5R531	0.3579	3.2983	0.1085	0.0010	0.1448	0.5711
ecf_5R532	0.9866	3.7013	0.2665	0.0002	0.4630	1.5101
ecf_5R535	0.8457	2.0858	0.4055	0.0374	0.0493	1.6422
ecf_5R539	0.4170	1.9488	0.2140	0.0518	-0.0033	0.8374

Table 14: District 5 LN_TASP Model II fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
594	563	36	0.5355	0.6723	0.6549	0.5213	0.4213	38.5036	0	975.904	1116.284

Table 15: District 5 LN_TASP Model II coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	7.3553	12.9170	0.5694	0.0000	6.2369	8.4738
lnsbldsfint	0.5485	6.6822	0.0821	0.0000	0.3873	0.7097
single_family1story	0.1548	3.2348	0.0479	0.0013	0.0608	0.2488
half_duplex1story	NA	NA	NA	NA	NA	NA
half_duplex2story	NA	NA	NA	NA	NA	NA
two_plus_baths	0.1789	2.4417	0.0733	0.0149	0.0350	0.3228
good	0.9102	10.3740	0.0877	0.0000	0.7378	1.0825
fair	-0.4447	-8.6899	0.0512	0.0000	-0.5453	-0.3442
poor	-0.5543	-7.0302	0.0788	0.0000	-0.7092	-0.3994
very_poor_unsound	-1.2288	-5.7057	0.2154	0.0000	-1.6518	-0.8058
qabove_avg	0.2545	2.3656	0.1076	0.0183	0.0432	0.4658
garage	0.1278	2.5775	0.0496	0.0102	0.0304	0.2251
fireplace	0.1050	1.9266	0.0545	0.0545	-0.0020	0.2121
elec_baseboard	NA	NA	NA	NA	NA	NA
wall	NA	NA	NA	NA	NA	NA
ecf_5R503	1.0586	5.3420	0.1982	0.0000	0.6694	1.4478
ecf_5R504	0.7111	5.8526	0.1215	0.0000	0.4725	0.9498
ecf_5R506	0.4567	1.7463	0.2615	0.0813	-0.0570	0.9704
ecf_5R507	0.2291	1.7079	0.1342	0.0882	-0.0344	0.4926
ecf_5R508	0.8029	2.4492	0.3278	0.0146	0.1590	1.4468
ecf_5R509	0.9869	5.0397	0.1958	0.0000	0.6023	1.3716
ecf_5R511	0.6568	3.1679	0.2073	0.0016	0.2496	1.0641
ecf_5R512	0.7084	3.8457	0.1842	0.0001	0.3466	1.0702
ecf_5R513	0.6078	2.5605	0.2374	0.0107	0.1416	1.0741
ecf_5R514	0.8814	10.5099	0.0839	0.0000	0.7167	1.0461
ecf_5R516	NA	NA	NA	NA	NA	NA
ecf_5R517	1.4694	5.9831	0.2456	0.0000	0.9870	1.9518
ecf_5R518	0.6210	1.8984	0.3271	0.0582	-0.0215	1.2635
ecf_5R521	0.4628	2.8730	0.1611	0.0042	0.1464	0.7793
ecf_5R524	2.6237	5.8688	0.4471	0.0000	1.7456	3.5018
ecf_5R529	0.9275	11.5234	0.0805	0.0000	0.7694	1.0856
ecf_5R530	0.9898	4.0479	0.2445	0.0001	0.5095	1.4702
ecf_5R531	0.3798	3.4895	0.1088	0.0005	0.1660	0.5935
ecf_5R532	1.0219	3.8766	0.2636	0.0001	0.5041	1.5397
ecf_5R535	0.8391	2.0589	0.4075	0.0400	0.0386	1.6396
ecf_5R539	0.3682	1.7685	0.2082	0.0775	-0.0407	0.7772

